



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.6A

Write and Evaluate Algebraic Expressions

Lesson 6-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can write an algebraic expression for a real situation and evaluate it for a given value of the variable.

Language Objective: I can explain my expression using the words variable, term, coefficient, constant, substitute, and evaluate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Write Algebraic Expressions

Variable

Algebraic Expression

Coefficient

Constant

Like terms

Substitute

Evaluate



Tap any word to see what it means and a picture.

1

Turn the words into numbers, operations and a .

2

A letter that stands for an unknown number is a .

3

A math phrase with variables, numbers, and operations but no equal sign is an

.

4

The number multiplied by a variable, like the 7 in $7n$, is the

.

5

A term that is just a number and never changes, like the 5 in $3x + 5$, is a

.

6 Terms with the same variable part are .

7 Replace the variable with a given number before you
 the expression.

8 To put numbers in for the letters and work out the value is to
 .

Write About the Math

Write an expression for the total cost. What does each part of the expression represent?

Escribe una expresión para el costo total. ¿Qué representa cada parte de la expresión?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Write Algebraic Expressions**

Escribir expresiones algebraicas

☐ **Variable**
Variable

☐ **Algebraic Expression**
Expresión algebraica

☐ **Coefficient**
Coeficiente

☐ **Constant**
Constante

Model: Key words name the operation, and 'less than' reverses the order. — A strong answer matches key words to operations (sum/more than = add, less than = subtract, product = multiply) and explains '12 less than g' starts with g, giving $g - 12$.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The total cost is** ____ **h. The** ____ **represents the** ____ **and** ____ **h represents the** ____.
- **My answer is** ____ **because** ____.

Mi respuesta es ____ *porque* ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** ____.
- **My evidence is** ____, **so** ____.

Mi afirmación es ____.

Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Write Algebraic Expressions
2. **Notes 2:** Variable
3. **Notes 3:** Algebraic Expression
4. **Notes 4:** Coefficient
5. **Notes 5:** Constant
6. **Notes 6:** Like terms
7. **Notes 7:** Substitute
8. **Notes 8:** Evaluate
9. **Watch for this mistake:** *A common mistake in Write Algebraic Expressions is treating 'each' or 'per' as if it means add instead of multiply, and attaching the variable to the wrong number. For 'a \$25 setup fee plus \$8 per hour, h,' students often write $25 + 8 + h$ (adding all three numbers) or $25h + 8$ (multiplying the flat fee by h) instead of the correct $8h + 25$. Since \$8 applies per hour, it must multiply the variable ($8h$), and the flat \$25 fee is a constant added only once — check that the per-unit rate, not the flat fee, is the number multiplied by the variable.*
10. **Try It 1:** A. $x - 5$ — 'Less than' subtracts from the number: start with x and take away 5, so $x - 5$.
11. **Try It 2:** A. $2s$ — 'Twice' means two times the number, so twice s is $2s$.